



Calibration and Comparison of Tumor Growth Models in the Dynamics of Murine Uveal Melanoma

Nicolly Baptista Alves¹, Paulo F. A. Mancera², Guilherme Rodrigues³.

Instituto de Biociências, UNESP, SP, Brasil

nicolly.baptista@unesp.br¹, paulo.mancera@unesp.br², g.rodrigues2@unesp.br³

Introduction

Melanoma, among different types of cancer, is known to be aggressive with a high potential for metastasis [1]. Mathematical modeling has been used to describe the dynamics of tumor growth and to investigate responses to different therapeutic approaches. Models based on ordinary differential equations (ODEs) allow to represent tumor temporal evolution as well as, to analyze biological parameters related to cell proliferation, tumor death, and therapeutic effects.

Objectives

- Calibrate and compare mathematical models of tumor growth using experimental data from untreated (control group) murine melanoma obtained from [2];
- Select the model which best describes each tumor lineage using statistical criteria;
- Evaluate the impact of outlier removal on model calibration and selection.

Methodology

Exponential, linear, logistic, surface, von Bertalanffy, Mendelsohn, and Gompertz growth models were evaluated to identify the one which best describes tumor growth dynamics in untreated mice. Model parameters were estimated using nonlinear Least Squares Method, while the numerical solution of the ODEs was performed with Python LSODA package. Model selection was based on the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC), which consider simultaneously the adjustment quality and the complexity of the model, in addition to other evaluation metrics. Experimental data from the NOG control group, consisting of immunodeficient mice lacking interleukin-2 (IL-2) expression, were analyzed for three tumor lineage. The MM33 lineage represents an immunotherapy-sensitive cutaneous melanoma, whereas 92-1 and UM121213B represent uveal melanoma corresponding to a patient-derived xenograft (PDX) established directly from the tumor tissue. In order to evaluate the influence of outliers on model calibration, the adjustment were compared between complete dataset and the obtained set after identifying and removing mice with markedly tumor volumes differed from the remainder of the group.

Results

In order to illustrate the impact of outliers on model calibration it was used UM121213B, among the three lineages analyzed, as it showed the fastest growth tumor within the group. During the analysis of the experimental data for this lineage, mice with faster tumor growth than the rest of the group were identified characterizing biological heterogeneity among individuals. To evaluate the influence of these points on model calibration, adjustments were compared using the complete dataset and the dataset obtained after removing these mice.

Table 1 shows the comparison of the BIC and RMSE adjustment indicators for the tumor growth models applied to the UM121213B lineage comparing adjustment obtained with the complete dataset and with the dataset after removal of the discrepant points obtained from [2].

Table 1: Model comparison — UM121213B: complete dataset vs. without outliers.

Modelo	BIC (with outliers)	RMSE (with outliers)	BIC (without outliers)	RMSE (without outliers)
Linear	1457,89	39,77	871,04	15,57
Gompertz	1447,92	37,69	876,07	15,60
Logístico	1498,55	45,78	876,63	15,87
Mendelsohn	1477,99	42,76	887,39	16,92
Von Bertalanffy	1471,95	41,84	907,91	18,89
Exponencial	1515,99	49,14	928,77	20,01
Superfície	1520,66	49,10	932,28	19,91

Figure 1 presents the models calibrated using the complete dataset extracted from the study by [2], allowing evaluation of adjustment performance in the presence of all data points.

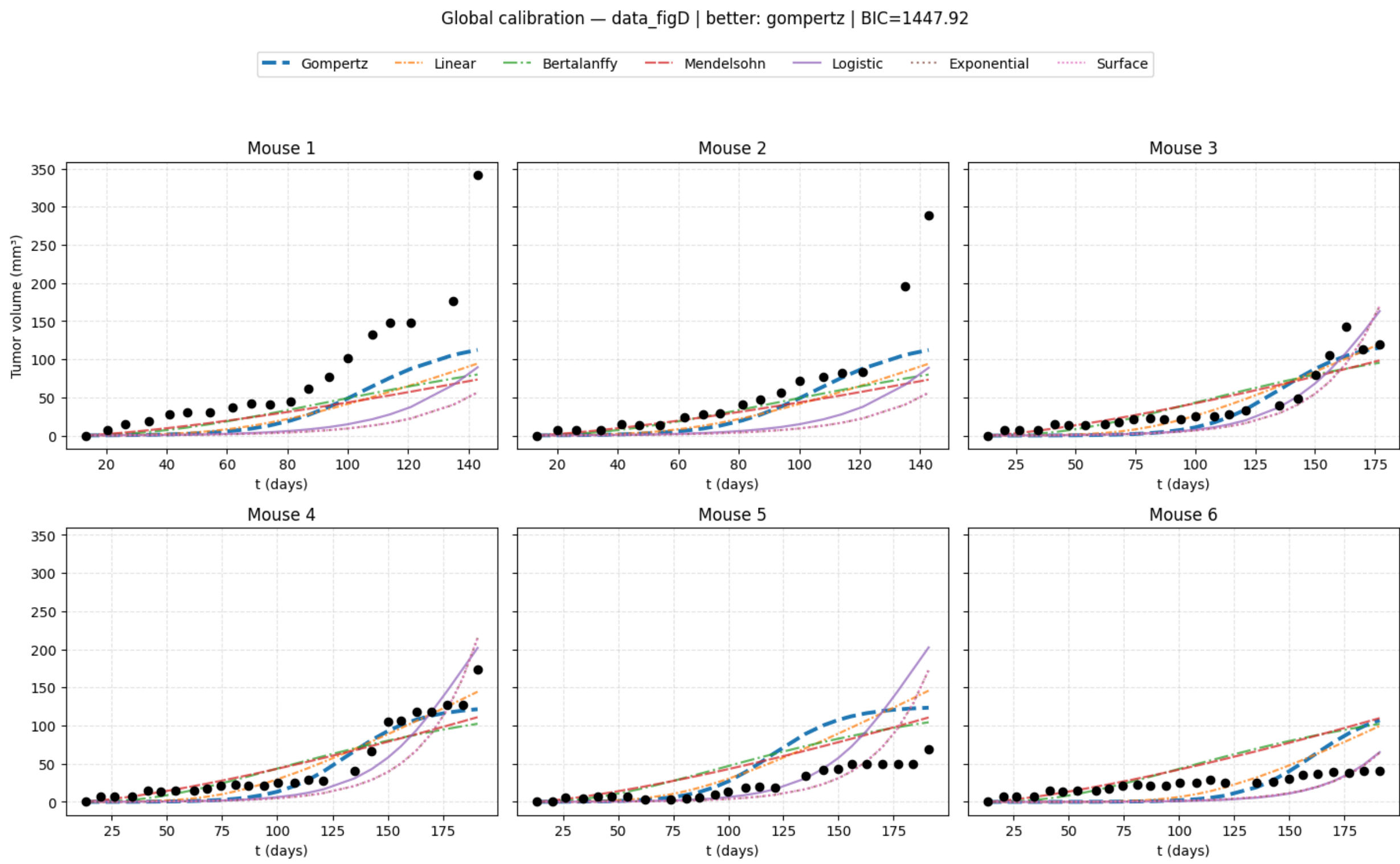


Figure 1: Model adjustments to the complete dataset (all extracted points). Source: Prepared by the authors.

Figure 2 presents a new calibration after removal of the identified outlier points, aiming to evaluate the influence of these points on the quality of the model adjustments.

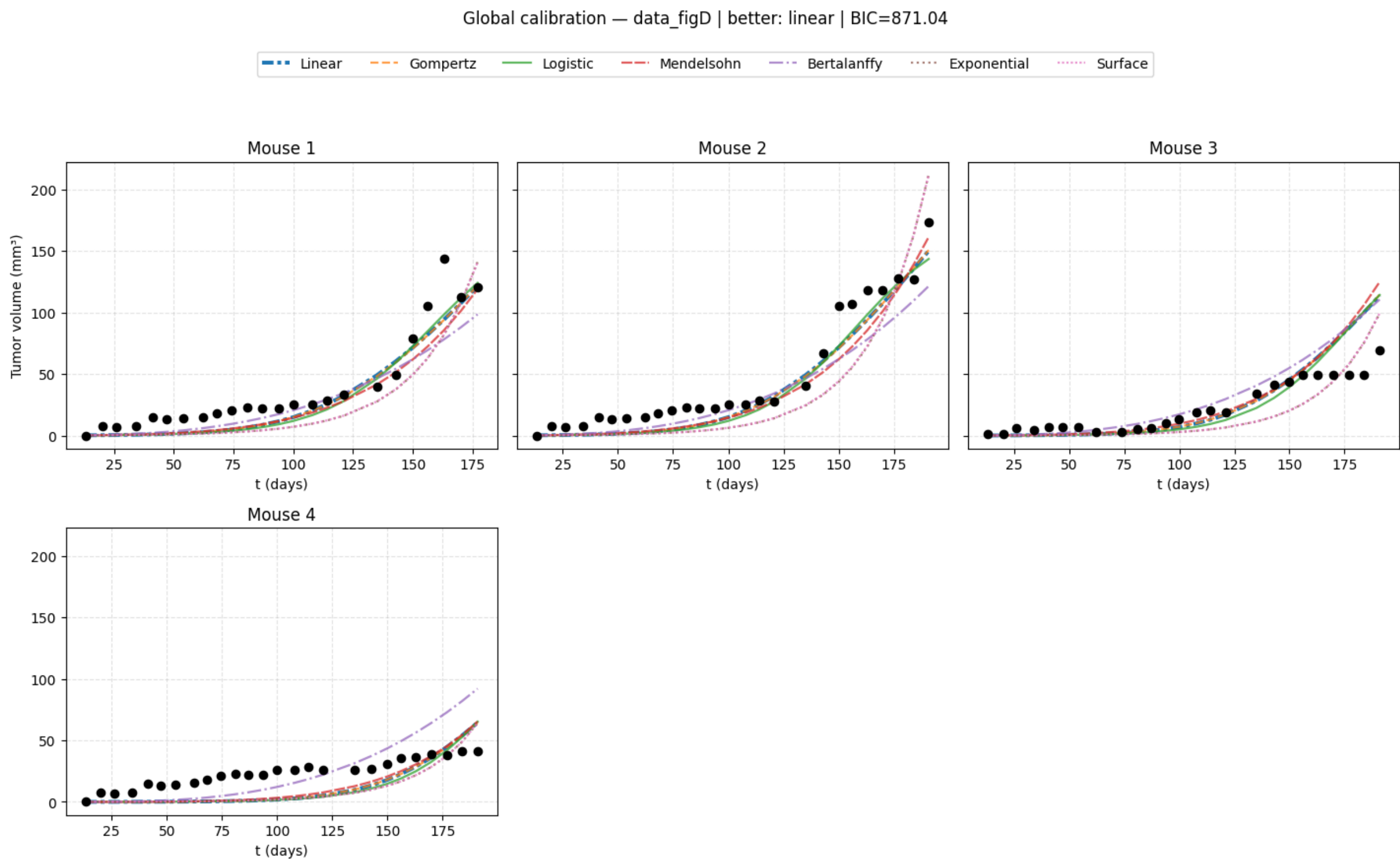


Figure 2: Model adjustments after outlier removal. Source: Prepared by the authors.

Conclusion

The calibration of growth models to the UM121213B lineage showed that the presence of outliers directly influences the selection of the most appropriate model: with the complete dataset the Gompertz model provided the best adjustment (BIC = 1447.92), whereas after removing mice with discrepant tumor growth the Linear model was selected (BIC = 871.04). This result demonstrates that model calibration is sensitive to the presence of discrepant values in the experimental data reinforcing the importance of careful outlier identification and treatment prior to model selection. The selected model will be fixed as a baseline to which a term representing the effect of treatment on tumor growth will be incorporated contributing to a quantitative understanding of the tumor response to CAR-T cell immunotherapy.

References

- [1] RODRIGUES, G.; SILVA, J. G.; ADIMY, M.; MANCERA, P. F. A. A mathematical model to the melanoma dynamics involving CAR T-cells. *Computational and Applied Mathematics*, v. 44, 2025. DOI: 10.1007/s40314-024-03060-3.
- [2] FORSBERG, E. M. V.; LINDBERG, M. F.; JESPERSEN, H.; ALSÉN, S.; BAGGE, R. O.; DONIA, M.; SVANE, I. M.; NILSSON, O.; et al. HER2 CAR-T cells eradicate uveal melanoma and T-cell therapy-resistant human melanoma in IL2 transgenic NOD/SCID IL2 receptor knockout mice. *Cancer Research*, v. 79, n. 5, p. 899–904, 2019.

Acknowledgments

We thank PIBIC/CNPq (Process N°. 20770) and UNESP for their support on this research.